

Kevin Liu Rodrigues

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I am a scientist with a PhD in statistical and computational physics and solid quantitative basis. During my masters, I started programming for interfacing with various electronic equipments, but, inspired by advances in artificial intelligence in the form of AlphaGo, in my PhD I shifted focus to statistical and computational physics.

During my Doctorate studies, I implemented large-scale dynamical systems simulations in C, which led to year-long residency at the University of Michigan. There, I developed mathematical frameworks that both explained and extended simulation results, and formally established properties of complex dynamical systems that had previously been understood only intuitively within the international research community.

This led me to a diverse experience in data science, machine learning and software development: In 2018 I joined Olivia AI to develop sentiment analysis and a chatbot before the advent of LLM's. By 2020 I was hired at Rappi to develop life time value models and user metrics to aid in the company's expansion in South America, and finally in 2021 I joined Spacetime Labs (now Moray AI) to work with machine learning for precision agriculture and crop management through climate and sensor data.

EXPERIENCE

Moray AI (2021 – Present) | *Data Scientist and Engineer*

- Machine learning and predictive models for precision agriculture and crop management.
- Data pipelines for gathering and processing spatio-temporal data from sources such as NASA, Copernicus, ECMWF, etc.
- Machine learning pipelines from training to deployment using mainly MLflow and AWS.

Rappi (2020 – 2021) | *Data Scientist*

- Development of user life-time value models and user metrics.
- Large dataset processing and visualizations including spatio-temporal maps and dashboards.

University of Michigan (2019) | *Assistant researcher*

- Development of large-scale simulations on dynamical systems and mathematical mean field descriptions.

Olivia AI (2018) | *Data Scientist*

- Development of a chatbot in the very early stages of AI, experimenting with sentiment analysis, string distance metrics and word embeddings.
- Classification models for text, including Neural Networks, SVM's and ensemble methods.

Federal University of Minas Gerais (2015 – 2016) | *Researcher and Engineer*

- Development of a high power near-field microscope for Hewlett-Packard (HP) under coordination of professor Gilberto Medeiros

Triu - College Prep Course (2012 – 2014) | *Volunteer math teacher*

- Math classes for low income students aiming to join top universities.

University of Campinas (2010 – 2012) | *Scientific initiation*

- Development of pico-Farad detection electronics for capacitive sensing.
- Development of electrostatic lenses for ion beam sampling and focusing.

EDUCATION

Doctorate in statistical physics | **UFMG** and **UMICH** (2016 – 2021)

(Advisors: Ronald Dickman and Kevin Wood)

Master of science in applied physics | **University of Campinas** (2012 – 2015)

Bachelor of science in physics | **University of Campinas** (2008 – 2012)

SKILLS

Programming languages: Python, SQL, C, C++, JavaScript

Tools: MLflow, Docker, Amazon Web Services (AWS), Airflow, Kubernetes, Parametric Modelling (CAD)

Languages: Portuguese (native), English (fluent spoken/written)